



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/AGSR Division

First Semester 2026-27
Course Handout (Part II)

Date: 03/08/2026

In addition to Part I (General Handout for all courses appended to the Time Table), this portion gives further specific details regarding the course.

1	Course No.	:	ME F216 & MF F216
2	Course Title	:	Materials Science and Engineering
3	Instructor In-charge	:	Sachin U Belgamwar
4	Instructors	:	Sachin U Belgamwar, Venkatesh K P, Sharad Shrivastava, Vandana Vidyadhari, Murali Palla
5	Practical Instructor	:	G. C. Mauli, Saket singh, Riya Sharma, Anuj Anil Deshpande

4. Course Description

Introduction, Structure of Materials (Metal and Ceramics), Dislocations, heat treatment of steel and strengthening Mechanisms of Metals, Phase diagrams, Iron-carbide phase diagram, Phase transformation in Metals, Mechanical and thermal properties of Metals, Polymers (Structure, processes and properties), powder metallurgy.

5. Scope and Objectives

CO1. Develop familiarity with the different levels of structure (atomic, crystal, microscopic) in engineering materials and deviations from “perfect” structure (structural defects).

CO2. Introduction to characterization method like optical microscope, XRD, SEM, TEM

CO3. Understand the effects of microstructure on the mechanical properties of materials.

CO4. Understand the basis for microstructure development in materials.

CO5. Understand how materials are processed.

CO6. Selection of Materials for specific application

6. Prescribed Text Book



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T1. Callister, William D., Materials Science and Engineering: An Introduction, 8th Edition, ISBN# 978-0-047-0419977, Wiley, 2010 5.

7. Reference Books

- R1. William F Smith, Javad Hashmi and Ravi Prakash, Materials Science and Engineering, Fourth Edition, Tata Mcgraw Hill Education Private Limited, New Delhi.
- R2. George E. Dieter, Mechanical Metallurgy, SI Metric Edition McGraw Hill Book Company, London.
- R3. R A Higgins, Applied Physical Metallurgy, Sixth edition, Viva Low priced students edition, New Delhi.
- R4. V. Raghavan, Material Science and Engineering

8. Course Plan

Module Number	Lecture session/Tutorial Session.	Reference	Learning Out come
1. Introduction to Engineering Materials, structure and properties.	L1.1.Introduction of Engineering materials, classification into metals, polymers and ceramics and their properties.	T 1.1-1.6.	Identifying the relationship between the material structure and its influence on Modulus of elasticity, melting point, strength, etc.,. This is useful in selection of materials.
	L 1.2. Atomic structure, crystal structure, micro structure and macro structure of materials.	T 2.1-2.7	
	T1.1. Exercise problems on structure property correlations.	T 1 and 2	
2. Basic crystalline structure of solids, miller indices for planes, directions, planer density, linear density and properties of solid influenced by crystal structure.	L2.1 Bravice lattice, unit cell, arrangement of atoms in common crystal structure	T 3.1-3.8.	Understanding the relationship between crystal structure and property. Effect of close packed plan and direction on slip system. Deformation behaviors of different crystalline materials.
	L2.2. Miller indices for planes and directions, identification of miller plan and directions. Determination of close packed plan, close packed directions, slip system and influence of crystal structure on properties of materials	T 3.8-3.16	
	T2. Exercise problems on crystal structure and its influence on properties.	T 3	
3. Crystal imperfections such as Point, line, planer and volume defects. Influence of crystal defects on properties of materials.	L3.1 Classification of crystal defects, point defects, vacancies, interstitials, vacancy concentration and influence of point defect on properties of materials.	T 4.1-4.6.	Diffusion process, grain boundary strengthening, ASTM grain size number. Effect of dislocation on formability.
	L3.2. Line defects-edge dislocation, screw dislocation, Burger vector. Influence of line defects on plastic deformation of	T 4.6-4.11	



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	materials and deformation by slip and Planer defects- tilt boundaries, twin boundaries, grain boundaries and surfaces. Influence of planer defect, strengthening mechanisms twinning,		
	T3. Exercise problems on crystal imperfections and its effect.	T 4	
4. Binary phase diagrams, isomorphous and eutectic systems. Cooling curves of pure metals, alloys, formation of phase diagram, Gibb's phase rule, lever rule and applications	L4.1 Cooling curve of pure metal and alloys, method of arriving at phase diagram from cooling curves, Gibb's phase rule and phase equilibrium.	T 9.1-9.5	Understanding the effect of phase diagram on microstructure and properties of materials and importance of alloy systems.
	L4.2. Isomorphous system of phase diagram with Cu-Ni system as example, liquidus, solidus line, lever rule and determination of phases. Eutectic system with Pb-Sn system as example, eutectic, hypo eutectic, hyper eutectic phase transformations, microstructure and properties	T 9.5-9.11	
	T4. Exercise problems on phase diagrams.	T 9	
5. Iron and Iron carbide Phase diagram- solidification of pure iron, allotropy, invariant reactions, steel and cast iron region. Different phases of iron-carbon system and their influence on properties.	L5.1 .Cooling curve of pure iron, allotropic transformations, effect of carbon. Three invariant reactions and regions	T 9.17	Microstructure of steel and cast iron. Influence of carbon on property of steel. Plain carbon steel and alloy steel and properties. Cast iron and it's applications.
	L5.2. Steel region of Fe-Fe ₃ C phase diagram, eutectoid reaction, phase transformation. Effect of Carbon on properties of steel, Plain carbon and alloy steels, their influence on microstructure and properties.	T 9.18	
	L 5.3 Cast iron region of Fe-Fe ₃ C phase diagram, white cast iron and it's properties, Grey, ductile and Malleable cast iron and their applications	T11.1-11.4	
	T5. Exercise problems on Iron- carbon system	T9 and T11	
6. Isothermal transformation, formation of TTT diagram, cooling rate and phase transformation, critical cooling rate, effect of alloying elements.	L6.1. Brief introduction to solidification of pure metals, critical nucleation sites and growth, transformation rate and temperature. Analogy between liquid to solid transformation and solid to solid	T 10.1-10.5	Equilibrium and non equilibrium cooling of plain carbon steel and alloy steel and its influence on property of





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Heat treatment of steel- Annealing, normalizing, hardening, tempering, austempering, martempering, size effect. Effect of heat treatment on microstructure and properties of steel.	transformation, S curves and formation of isothermal transformation diagram, Metallic glasses.		steel. Influence of CCT on casting, welding and other manufacturing processes. Making of tool steels. Cost associated with alloy steel and decision making.
	L6.2. Formation of TTT diagram for eutectoid steel, different phases and regions, influence of cooling rate and phase transformation, critical cooling rate and martensitic transformation.	T 10.6-10.9	
	L 6.3 Size effect, hardenability, effect of alloying elements, Austempering and martempering .	T11.8	
	T6. Exercise problems on TTT and CCT diagram.	T 110 and T11	
7. Mechanical properties of materials, tensile testing, fatigue, creep and fracture of materials.	L7.1. Stress-strain diagram of metals, polymers and ceramics. Engineering stress, engineering strain, elastic modulus, True stress, true strain, and relationship.	T 6.1-6.11	Understanding the failure of materials and preventing the catastrophic failure, life estimation and residual life assessment. Historic failures and development of fracture mechanics.
	L7.2. Ductile and Brittle fracture, fractography, impact testing, ductile to brittle transition temperature and failure of materials.	T 8.1-8.6	
	L 7.3 Fatigue and creep, SN Curve, endurance limit, Types of fatigue cycle and fatigue life estimate..	T8.7-8.11	
	L 7.4 Creep, typical creep curve, effect of temperature and stress on creep curve, stress rupture test and determination of creep life. LMP approach.	T8.11- T8.15	
	T9 and T10- Tutorial problems on fatigue, creep and fracture.	T6,T8	
8. Structure of Ceramics, types of crystals, Determination of density. Applications and properties of Ceramics.	L 8.1 Determination of coordination number AX, AmXp and AmBnXp structures of Ceramics.	T 12.1- 12.3.	Structure and application of ceramics. Modern ceramic materials and phase diagram.
	L8.2. Density Calculation, silicates and polymorphs of Carbon and their structure. Types and application of ceramics.	T 12.4-12.9 T 13.1- 13.7	
	T 11. Tutorial Problems on Ceramics.	T 12,13.	





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9. Structure of polymers, types, application and properties.	L 9.1Determination of coordination number AX,AmXp and AmBnXp structures of Ceramics.	T 14.1-14.7	Structure and application of polymers. Modern polymeric materials.
	L9.2 Polymeric structure, crystallinity, polymeric molecules and chemistry, determination of molecular weight. Glass transition temperature, Properties and application of polymers. Advances polymeric materials	T14.8-14.12	
	T 12. Tutorial Problems on polymeric materials	T 14	
10. Composite materials.	L10.1Types of composite materials, rule of mixtures and it's applications.	T 16.1-16.6	Understanding of FRP, MMCs, CMC etc., and selection for different applications and tailor made materials. FGMs and applications.
	L10.2.Control of composite properties, different types and applications, Manufacture and testing of composites	T16.7-16.1	
	T13. Tutorial problems on Composite materials.	T16	
11. Selection of Materials.	L11.1.Materials chart with properties.	T 22.1-22.6	Understanding the selection of materials using materials performance index. Life, cost and properties base selection of real time components.
	L11.2Determination of materials performance index and it's use in materials selection decision. Application of materials selection with case study.	T22.6-22.14	
	T14. Tutorial problems on selection of Materials.	T22	

Lab demonstration experiments and case studies. (Optional)

- 1) Measurement of Young's Modulus, Tensile test on Mild Steel
- 2) Measurement of Impact Test
- 3) Measurement of Hardness
- 4) Fatigue testing and observation of fractured surface.
- 5) Measurement of shear modulus by torsion test on mild steel
- 6) Photoelasticity
- 7) Measurement of Critical buckling load
- 8) Rotating-Bending fatigue testing



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9. Evaluation Scheme:

Evaluation components.	Weightage	Day, Date, Session, Time
Evaluative Tutorial	10% (Best 04 out of 06)	Will be announced in the class
Surprise test in Tutorial	10%	Surprise
Mid semester examination.	90 Minutes (25%)	CB
Comprehensive examinations	180 Minutes (35%)	CB+OB
Laboratory Experiments and record	10%	OB
Laboratory test	10%	CB

After completing this course the students will be able to

- 1) Select suitable material for the specific application subjected to different constraints in terms of cost, availability, properties, life etc.
- 2) Understand the relationship between the structure and properties of materials.
- 3) Select suitable heat treatment process of steel to get the desired microstructure and property.
- 4) Identify the materials from its microstructure.
- 5) Understand the type of fracture from the fractured surface.

Closed Book Test: No reference material of any kind will be permitted inside the exam hall.

Open Book Exam: Use of any printed / hand written reference material (books and notebooks) will be permitted inside the exam hall. Loose sheets of paper will not be permitted. Computers of any kind will not be allowed inside the exam hall. Use of calculators will be allowed in all exams. No exchange of any material will be allowed.

The passing marks for this course will be determined based on one of the two criteria: 30% of the median of the post-comprehensive examination score or 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as passing marks in this course.

Consultation hour: To be announced during the lecture.



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Notices: All notices concerning the course will be uploaded on the **Nalanda-aws** portal **or will be shared by email.**

<https://nalanda.bits-pilani.ac.in/>

Attendance Policy:

Each student is expected to attend all classes, tutorials, and practical sessions and to participate in all evaluation components. A 10% weight is assigned to assessment linked to attendance and participation. To this end, a small surprise quiz will be conducted in the tutorial class. The scores in surprise quizzes linked to attendance and participation will be considered in the final score. This score will be considered only for students with a minimum attendance of 50% in the lecture and tutorial classes separately. For students with attendance less than 50% in either the lecture or tutorial class, zero marks will be assigned, irrespective of the scores in the surprise quizzes attempted.

Note:

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend lectures and the lab demonstration as per the schedule announced in Nalanda. Mid Semester Test and Comprehensive Examination according to the Evaluation Scheme given in the respective Course Handout. If the student is unable to appear for the Regular Test/Examination due to genuine exigencies, the student must refer to the procedure for applying for Make-up Test/Examination. **No make up for the tutorials.**

(Sachin U Belgamwar)

Instructor In charge

ME F216 & MF F216.



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